ALAGARSAMY DURAIKUMARAN | Singaporean | GitHub | LinkedIn

EDUCATION

Nanyang Technological University (NTU), Singapore Aug 2024 - May 2028
Bachelor of Engineering (Honours) in Computer Engineering
Relevant Modules - Algorithm Design and Analysis, Software Engineering, Data Structures and Algorithm

WORK EXPERIENCE

TATA Consultancy Services (TCS), Pace Port Singapore May 2026 - Present
Generative AI Engineer Intern

- Authored a quantum computing research report on hybrid quantum-classical transformer approaches and presented to the internal team.
- Built a Voice AI agent on Google's Gemini model, and with MCP and webhook integrations plus a lip-syncing avatar frontend, enabling real-time conversational interaction for client-facing use cases.
- Designed and delivered a Management Based Game with direct client interaction, simulating real-world business decision-making in an AI-augmented environment.
- Co-planned a large-scale AI workshop for 500+ students with a government partner.

SKILLS

- AI/ML: LLMs, LangChain, LangGraph, RAG pipelines, Google ADK, Ollama
- Frontend: React, Vite, Tailwind CSS, JavaScript, HTML, CSS
- Cloud & Backend: Python, Java, SQL, AWS, Supabase, Node.js, Flask

PROJECTS

Personal Project - SG Lotto Pro | GitHub | Demo Video Dec 2025 - Jan 2026

- Engineered end-to-end system leveraging Python, Flask, OpenCV, Tesseract OCR, Selenium, SQLite, and Bootstrap, enabling image parsing, automated web scraping, analytics, and persistent ticket history.
- Automated Singapore TOTO/4D result checking to save users 30 minutes per draw, allowing character recognition on mobile images.

Personal Project - AI ATS Agent | GitHub | Demo Video Nov 2025 - Dec 2025

- Developed an AI-powered resume analysis tool and interactive Streamlit web application leveraging Python and prompt-based LLM workflows for real-time document comparison.
- Reduced manual screening time by 60% by detecting skill gaps and missing keywords while providing structured feedback to improve resume-job match accuracy.

SAP x NTU PEAK Program (Industry Project) Aug 2025 - Oct 2025

- Selected for NTU PEAK industry program with SAP and proposed SAP Matrix, a solution concept integrated with Joule AI Agent.
- Delivered recommendations reducing carbon footprint estimation errors by 15% and material waste by 20% through AI-driven predictive logistics and supplier auditing.

HACKATHONS

Deep Learning Week 2026 Hackathon (Finalist) - CrashGuard AI | GitHub | Demo Video Feb 2026 - Mar 2026

- Built a privacy-preserving emergency intelligence web application using React, JavaScript, Python, TensorFlow Lite, and Ollama, enabling real-time crash detection, AI incident analysis, and automated emergency reporting through an end-to-end vision and LLM pipeline.
- Improved emergency response readiness by generating structured crash assessment reports and automatically notifying emergency contacts/dispatchers via EmailJS, reducing manual reporting time during accidents.

Microsoft x NTU Product Club Hackathon (2nd Place) – PromptHub

- Built PromptHub, an AI prompt marketplace developed during the Microsoft x NTU Product Club hackathon, enabling users to create, discover, and sell high-quality prompts for LLM workflows.
- Improved prompt accessibility and monetisation by implementing searchable prompt libraries and creator listings, reducing time spent crafting effective prompts for AI tools.

LEADERSHIP EXPERIENCE

NTU Students' Union, NTUSU Sep 2024 - Aug 2025
CCA Liaison Director

- Coordinated 200+ CCAs and streamlined operations while leading the annual Welcome Week for 5,000+ students, enhancing campus engagement.